

Genetic Architecture and Risk Prediction of Gestational Diabetes Mellitus in Chinese Pregnancies

Corresponding Author: Professor Siyang Liu

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This study explores an important and relevant issue: the early prediction of gestational diabetes mellitus (GDM), based on genetics. Using a non-invasive prenatal test (NIPT), the authors develop a prediction model that can identify at-risk individuals before 20 weeks of gestation. This application of readily available genetic data to GDM prediction could potentially be a game-changer in the clinical setting. The primary strength of this study is the large sample size. The studied cohort of (N = 116,144) pregnant women substantially exceeds the size of previous East Asian GDM studies. In conjunction with the identification of novel loci (14 for GDM and 105 for five other glycemic traits), and the integration of genetic and clinical data for a population underrepresented in GDM genetic research, I believe this paper presents a significant advance in the field of gestational diabetes prediction.

Major comments:

- The authors need to compare to existing work. Specifically, they should show the relative performance of the proposed model compared to existing models, including those based on clinical data alone and those incorporating limited genetic information.
- While the authors attribute observed discrepancies in genetic effects observed between the study and other datasets to power limitations in the MAGIC study, a more in-depth analysis of potential contributing factors, including population stratification, phenotypic heterogeneity, and methodological differences, should be conducted.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

Gu et al have preformed a study with the intent to predict gestational diabetes in early pregnancy. My comments is as following:

1. The authors preformed their GWAS using plink and I question why not more sophisticated methods taking relatedness into account (such as BOLT-LMM, REGENIE etc) was used when even these methods are available. Using a GRM in the GWAS could account for residual confounding.
2. The GWAS was adjusted for heritable confounders, such as BMI, which could be problematic and I would suggested that the authors also perform the GWAS without these adjustments to compare the results.
3. I could not find any QQ plots or lambda or LD score intercept values to assess the genomic inflation in the paper, and I think this needs to be added to allow us to properly assess the quality of the GWAS results.
4. The authors report to have identified 195 loci significantly associated with five glycemic traits. Are these independent signals? I feel information are lacking in the results as to how to interpret the data.
5. The authors should address the power differences in T2D and GDM GWAS as some of the differences could be due to size differences between studies.

6. Prediction models: As far as I can see the AUC confidence intervals between model 1-3 all overlap - how can the authors conclude that adding these genetics factors is actually better?

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

This is a manuscript by Yuqin Gu et al. reporting a study that used non-invasive prenatal screening to build PRS and combined with biomarkers measured <20 weeks of gestation for prediction of GDM dx at 24-28 weeks in large population of pregnant women in China.

The study has multiple strengths, including large sample size, leveraging two hospital-based samples, with similar rate of GDM (14-15%), using NIPT genotyping data + 43 biomarkers, performing GWAS of multiple glucose traits in addition to GDM diagnosis, and used of recent and appropriate statistical approaches for genetic analyses (e.g. for building PRS, and for comparison of effect size for pregnancy versus outside of pregnancy)

My concerns are mostly regarding interpretation and conclusions put forward by the authors. First, they should acknowledge that this approach may not be applicable to other settings where NIPT does not offer standard genome sequencing or large-scale genotyping. Second, their stated conclusions are a bit strong, the most informative marker is still glucose – in this current study measured at 16 weeks, to predict GDM at 24 weeks (yes, glucose predicts glucose).

I also have a few additional specific comments below:

Methods:

How were excluded patients with pre-gestational diabetes mellitus (PGDM)?

Were A1c at 12 weeks, and/or FPG at 16 weeks used to 'exclude' pre-gestational diabetes? If so, what thresholds were used?

The authors should be clearer in their methods about the difference between fasting glucose (I think meaning measured at 16 weeks of gestation) and glucose at time = 0 for the OGTT (which is also fasting but measured at 24-28 weeks)

In figure 1 and figure 5, can the authors include the population (sample size) used of the predictive analyses (XGBoost)

Results:

The authors stated that the fasting glucose was measured in first trimester (line 240) but in Supp Table 1 – it's indicated mean gestational age 16 weeks

GDM and T2D in East-Asian: the authors found that MTNR1B is shared but the visual inspection of their own Figure 4A (and prior knowledge from European GDM versus T2D) suggest GDM predominant – how do the authors explained the discrepancies?

In line 294: I think the authors meant Table S1:

and 43 biomarkers and electronic health record phenotypes listed in Table S2

In lines 300-302: please add the gestational weeks at which the FPG and A1c was measured for clarity

The first model using 46 clinical features achieved an AUC of 0.700 [0.643-0.745], and an accuracy of 0.821 [0.803-0.827] (Figure 5A, Table S8), with FPG (11.8%), maternal age (6.57%), and HbA1c (4.33%) as the top contributors (Figure S8)

Discussion: the authors mis-reported their own findings in lines 350-352

In Figure 5, it's the PRS model that has an AUC =0.630 (not the phenotypic model)

From the Discussion text: This combined model achieved an AUC of 0.758 [0.682-0.768], and an accuracy rate of 0.830 [0.807-0.837] before 20 weeks of gestation, significantly outperforming models using only phenotypic data, which had with an AUC of 0.630 [0.559-0.654] and an accuracy rate of 0.803 [0.772-0.810].

Discussion: could the authors expand about how they would use the predicting algorithm? If they identify the 0.1% of pregnant women at 'high risk' of GDM (based on the cut-off for OR at 1.65) would they target these women by asking to do an OGTT earlier than the standard 24 weeks? Or offer them right away monitoring blood glucose and treatment?

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

All of my comments have been addressed to my satisfaction.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

I appreciate the authors for their response to my concerns. I am however not quite satisfied with some of their responses.

1. I suggested using a software like REGINE or BOLT-LMM to handle population stratification in the sample. Just removing related individuals is not enough to mitigate the effect of population stratification. The authors response is with concerns regarding SAIGE. I have myself seen these issues with SAIGE and would not recommend this software, however, the field standard is to use a software with a GRM when that is possible, and I do think the authors should reconsider using a best practice tool for their analysis.

2. The authors have changed their reporting on new loci to state that they identified 19 independent loci. To me it is still not clear from their description in the methods how many of these are novel. Have they been associated with GDM in other populations? Or Type 2 Diabetes? The authors state that they were not found in "European GDM GWAS finding", but the paper is not referenced and there are several papers published that should be checked. Further on this they state that 205 independent signals were associated with the five glycemic traits. I still wonder if it is in fact 200 new loci or is it 40 loci in each of the 5 GWAS? And are they novel? Or previously known GDM, glucose or T2D hits?

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The authors have addressed my comments appropriately

(Remarks on code availability)

Version 2:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

I thank the authors for taking onboard the comments and addressing some of them in a rigorous way. I still don't 100% agree with their rationale for using plink2, but I am glad they have shown that it for the main is showing concordant results as for BOLT or REGENIE.

For my second point of my last review I stated that I am not clear on how many findings are actually novel, and I still am not based on the response from the authors. To me it reads as inflation of the results because the authors have performed 5 GWAS of glycemic traits in addition to GDM. All these traits are highly correlated and when I look at table S4 which list all the SNPs and their genes I see that many of the genes are present in more than one GWAS. For example variants in CDKAL1, PCSK1, MTNR1B to name a few are listed over and over again. It sounds very impressive when the authors write 205 loci, but reality is seemingly a lot fewer new genes have been associated with any of these traits.

The authors write in the response: "we clarify that the association findings are phenotype-specific. We did not merge loci across different traits" and I wonder why you would not consider Fasting glucose at two time points to be the same or similar phenotype and cross check if the same gene pop up in both these GWASs?

To me this reporting feels inflated and not accurate, and perhaps this is also reflected in the title quoting an impressive "116,144 Chinese Pregnancies" when the individual GWASs have a sample size ranging from 56,000 to 85,000?

(Remarks on code availability)

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Manuscript Title: Integrating Genetics for Early Prediction of Gestational Diabetes Mellitus: Insights from Non-invasive Prenatal Sequencing of 116,144 Pregnant Women in China

Reviewer Comments & Author Rebuttals

We would like to express our gratitude to the editors and reviewers for their constructive feedback and valuable recommendations. We have carefully addressed each question and present our point-by-point responses below. The revised contents are highlighted in blue text in the revised manuscript, and the changes are quoted in italics within this response.

In response to the reviewer's comments, we have enhanced the modelling approach by incorporating an additional dataset. In the initial version of the manuscript, the prediction model was trained and evaluated using data from 3,565 participants in the NIPT-PLUS cohort. To improve the evaluation of our machine learning model, we have now included the Baoan 20K cohort, which consist of 20,439 participants who underwent NIPT at Baoan Hospital after 2021. Specifically, the model was trained using 60% of the Baoan 20K cohort data, tested on 20% of the cohort, and validated on the remaining 20%. Additionally, we validated the model externally using the NIPT PLUS cohort, which provides increased NIPT sequencing depth. This expanded train-test-validation design ensures a more robust evaluation of the model's performance and its generalizability to other hospitals using NIPT data. The details of this methodology are provided in Figure 2 and described in the Methods section of the revised manuscript.

The key conclusions of the study remain unchanged. Our analysis continues to show that incorporating genetic information from polygenic risk scores (PRS) of GDM and glycemic traits significantly improves the prediction models that rely solely on phenotypic data (DeLong's test $P < 0.05$), and we found that PRS for GDM, OGTT2H and OGTT1H emerged as crucial predictors in the prediction model.

Furthermore, in accordance with the formatting requirements instructed by the editors, we have revised the manuscript title from the original "Integrating Genetics for Early Prediction of Gestational Diabetes Mellitus: Insights from Non-invasive Prenatal Sequencing of 116,144 Pregnant Women in China" to "Genetic Architecture and Risk Prediction of Gestational Diabetes Mellitus in Over 116,144 Chinese Pregnancies".

Author Rebuttals to Initial Comments:

Referees' comments:

Referee #1 (Remarks to the Author):

This study explores an important and relevant issue: the early prediction of gestational diabetes mellitus (GDM), based on genetics. Using a non-invasive prenatal test (NIPT), the authors develop a prediction model that can identify at-risk individuals before 20 weeks of gestation. This application of readily available genetic data to GDM prediction

could potentially be a game-changer in the clinical setting. The primary strength of this study is the large sample size. The studied cohort of (N = 116,144) pregnant women substantially exceeds the size of previous East Asian GDM studies. In conjunction with the identification of novel loci (14 for GDM and 105 for five other glycemic traits), and the integration of genetic and clinical data for a population underrepresented in GDM genetic research, I believe this paper presents a significant advance in the field of gestational diabetes prediction.

Major comments:

- The authors need to compare to existing work. Specifically, they should show the relative performance of the proposed model compared to existing models, including those based on clinical data alone and those incorporating limited genetic information.

We thank the reviewer for this valuable suggestion. In response, we conducted a thorough comparison with existing literature on the early prediction of GDM. A PubMed search was performed for studies published in English between Jan 1, 2019, and Jul 1, 2024, using the search query used: ((Gestational diabetes [MeSH Terms]) AND (Prediction)). This search identified 15 relevant studies on GDM prediction, which are summarized in Table S19 of the revised manuscript.

Of these studies, 13 used only clinical records for prediction, with area under the receiver operating characteristic curve (auROC) values ranging from 0.732 to 0.946. The study with the largest same size to date reported an auROC of 0.710, which is comparable to the performance of models based solely on clinical phenotypes in our study. Two studies incorporated genetic data: one study involving 962 participants in China (Qi Wu et al.) reported an auROC of 0.620 using clinical records and 4 SNPs in the validation dataset; the other, involving 5,085 Chinese participants (Yipeng Wang et al.), reported an auROC of 0.960 using copy number variants, but this study had significant limitations, including a lack of external validation and a small sample size.

Compared to these studies, our integrated model achieved auROC values of 0.729 for the test data set, and 0.729 and 0.710 for the two independent external validation datasets. This performance significantly outperformed models based solely on clinical records in our study (DeLong's test $P < 0.05$).

We have included these comparisons in the discussion section of the revised manuscript (Lines 290 to 295) as follows:

“Compared to earlier studies on early GDM prediction that incorporated genetic data, one study involving 962 participants in China reported an AUC of 0.620 using clinical records and 4 SNPs associated with T2D¹¹. Another study, involving 5,085 Chinese participants, achieved an AUC of 0.960 using copy number variants but it lacked external validation²⁸(Table S19). Our study demonstrates superior performance, reliability, and robustness in predicting GDM risk.”

	Our study	MAGIC
Population	Chinese	South Korean
Age[Mean (SD)]	women 29.6(4.2)	men 51.8 (8.8) / women 52.6 (9.0)
Sex	female	Male and female
Imputation software	glimpse	minimac
Software for GWAS analysis	plink2	Mach2QTL V1.1.3
Sample size	84,743	8,509
Covariates	PC1-PC10, BMI, Age, OGTT gestational week	Recruitment area, sex, age, PC1, PC2, PC3, BMI

Reviewer #2 (Remarks to the Author):

Gu et al have preformed a study with the intent to predict gestational diabetes in early pregnancy. My comments is as following:

1. The authors preformed their GWAS using plink and I question why not more sophisticated methods taking relatedness into account (such as BOLT-LMM, REGENIE etc) was used when even these methods are available. Using a GRM in the GWAS could account for residual confounding.

We appreciate the reviewer for the suggestions. In our previous analyses, we observed that linear mixed model (LMM) implemented in SAIGE (PMID: 30104761) generates lambda values less than 1 for certain phenotypes (Figure R2). The underlying reason for this remains unclear. Therefore, we are using our pipelines (PMID: 39389018) for the analysis: we first excluded related participants from the cohort and subsequently applied logistic or linear regression models using PLINK, incorporating principal components along with maternal age, BMI, and gestational week as covariates. Judgement of accuracy of the analytical results were determined in the replication datasets.

Notably, the GWAS results obtained using LMM in SAIGE were highly consistent with those generated by the regression model in PLINK, yielding comparable estimates of genetic effects. To illustrate, we conducted a comparison of GWAS results for GDM using SAIGE. The findings demonstrated strong concordance with the PLINK analysis, with all lead SNPs for GDM showing *P-values* below the genome-wide significance threshold of 5×10^{-8} in the SAIGE results. Figure R2 and Table R2 provide a summary of these comparisons.

Considering the above information, we would like to retain the PLINK results in the manuscript.

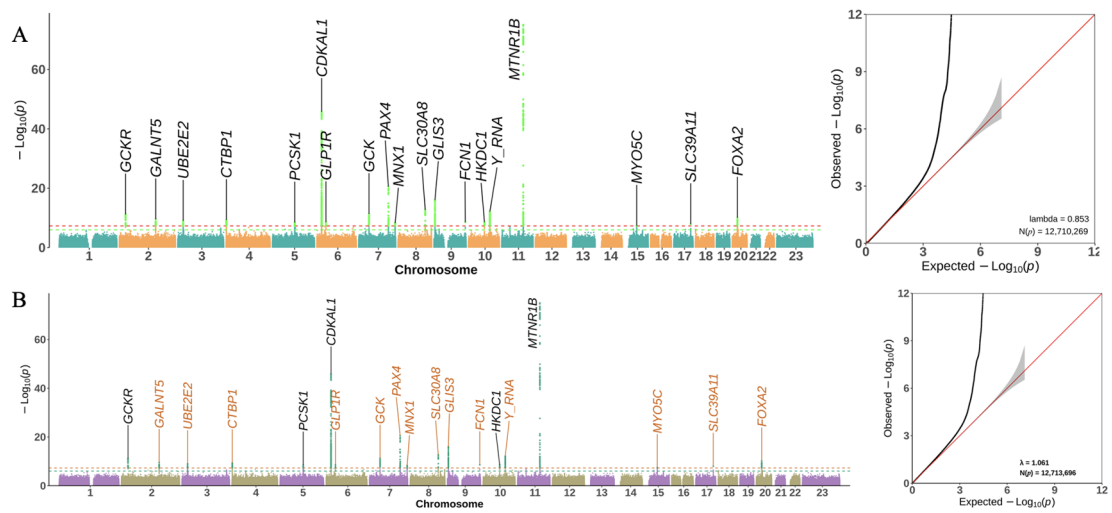


Figure R2. Comparison of GWAS results for GDM between PLINK and SAIGE analyses.

Panel A and B present the results of the GDM GWAS conducted using SAIGE and PLINK, respectively. The slightly lower number of variants in the SAIGE results (12,710,269) compared to the PLINK results (12,713,696) is attributed to differences in variant filtering criteria in SAIGE where variants with $MAC < 10$ were filtered.

Table R2: Comparison of the effects of GDM lead SNPs between PLINK and SAIGE analyses.

SNP	CHR	BP	A1	A2	FRQ	BETA	SE	P	GENE	BETA_SAIGE	SE_SAIGE	P_SAIGE
rs11127048	2	27529596	A	G	0.433	0.138	0.020	6.21E-12	GCKR	0.138	0.020	6.18E-12
rs12694904	2	157290112	T	C	0.144	0.179	0.028	2.91E-10	GALNT5	0.179	0.028	2.77E-10
rs76435632	3	23590683	C	G	0.858	-0.144	0.024	9.34E-10	UBE2E2	-0.145	0.024	8.80E-10
rs73069940	4	1242714	C	G	0.669	0.108	0.017	4.39E-10	CTBP1	0.108	0.017	4.48E-10
rs2882298	5	96391471	A	G	0.665	0.105	0.018	3.28E-09	PCSK1	0.105	0.018	3.36E-09
rs9368222	6	20686765	A	C	0.408	0.256	0.018	1.71E-46	CDKAL1	0.256	0.018	1.62E-46
rs9366994	6	39080284	T	C	0.300	-0.140	0.023	2.39E-09	GLP1R	-0.140	0.023	2.47E-09
rs741037	7	44193234	A	G	0.206	0.193	0.028	4.85E-12	GCK	0.193	0.028	4.59E-12
rs61160304	7	127609605	T	C	0.081	0.353	0.037	3.06E-21	PAX4	0.354	0.037	2.76E-21
rs4716672	7	156993814	T	C	0.751	0.134	0.023	5.18E-09	MNX1	0.134	0.023	5.35E-09
rs9650069	8	117191781	T	C	0.429	-0.202	0.028	1.80E-13	SLC30A8	-0.202	0.028	1.83E-13
rs10758593	9	4292083	A	G	0.409	0.185	0.022	9.57E-17	GLIS3	0.185	0.022	9.47E-17
rs7851502	9	134960660	A	G	0.562	-0.144	0.024	2.58E-09	FCN1	-0.144	0.024	2.56E-09
rs4746822	10	69223185	T	C	0.260	0.154	0.026	2.48E-09	HKDC1	0.154	0.026	2.40E-09
rs10882102	10	92706738	C	G	0.825	-0.146	0.020	7.26E-13	Y_RNA	-0.146	0.020	6.90E-13
rs10466351	11	92964815	T	C	0.431	0.449	0.024	6.65E-78	MTNR1B	0.448	0.024	6.14E-78
rs4776024	15	52237929	A	C	0.703	0.111	0.020	3.97E-08	MYO5C	0.110	0.020	4.11E-08
rs58299944	17	73011782	T	C	0.799	0.219	0.038	1.05E-08	SLC39A11	0.219	0.038	1.11E-08
rs6113722	20	22576461	A	G	0.143	-0.156	0.024	6.99E-11	FOXA2	-0.154	0.024	8.08E-11

2. The GWAS was adjusted for heritable confounders, such as BMI, which could be problematic and I would suggested that the authors also perform the GWAS without these adjustments to compare the results.

We appreciate the reviewer's suggestions. In our study, the GWAS was conducted using the formula GDM/glycemic traits ~ genotype + covariates (Age + PC1-10 + gestational week + BMI). To address the reviewer's concern regarding adjustments for heritable confounders, we performed additional GWAS analyses excluding BMI as a covariate.

Our results showed no significant differences between the analyses that included BMI as a covariate and those that excluded it (Figure R3, corresponding to Figure S2 in the revised manuscript). BMI was included as a covariate in our primary analyses because we aimed to investigate BMI-adjusted GDM disease liability. In addition, the results excluding BMI as a covariate have been included as secondary analyses in the revised manuscript for completeness upon the reviewer's suggestions.

In the revised manuscript, we added the following to the Methods section (Lines 386 to 387):

"As a secondary analysis, we also performed GWAS excluding BMI as a covariate to assess its impact as a heritable confounder."

And to the Results section (Lines 97 and 98):

"Excluding BMI as a covariate yielded consistent results (Figure S2A, Table S2)."

Additionally, in Lines 122 to 123:

"The QQ plot and LDSC intercepts indicated negligible statistical inflation (Figure S1), and results excluding BMI as covariates were also highly consistent (Figure S2B, Table S2)."

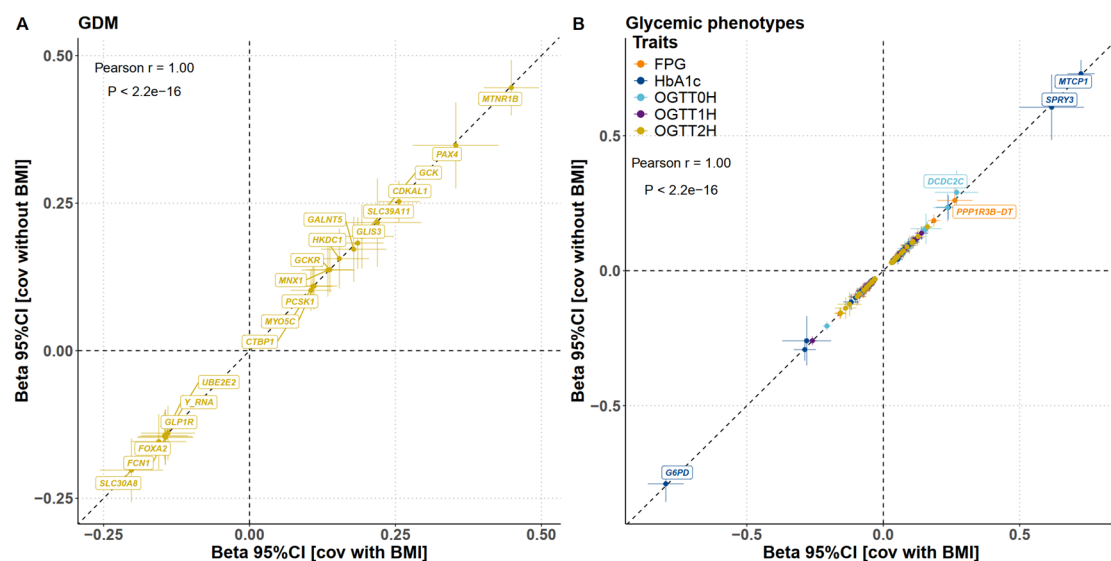


Figure R3. Comparison of lead SNP effects between primary GWAS and secondary analysis excluding BMI as a heritable confounder.

- I could not find any QQ plots or lambda or LD score intercept values to assess the genomic inflation in the paper, and I think this needs to be added to allow us to properly assess the quality of the GWAS results.

We appreciate the reviewer's suggestions. In the revised manuscript, we have added Figure S1 to summarize these results, which are also presented below in Figure R4.

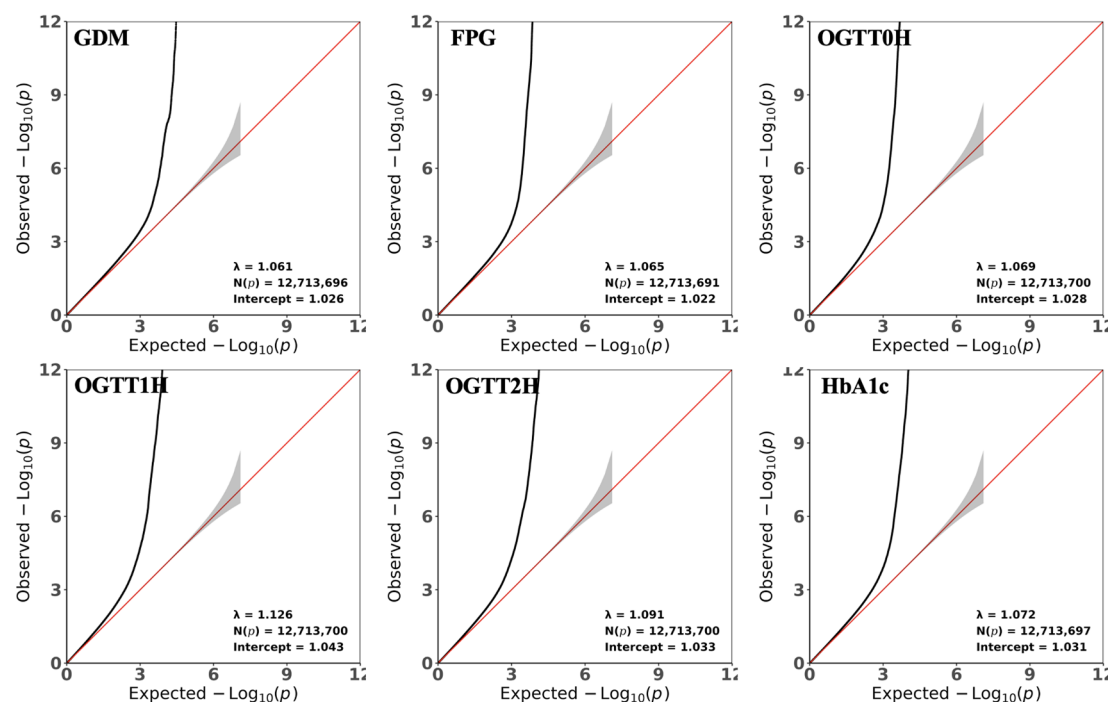


Figure R4. QQ plot of GWAS results for GDM and five glyceic traits.

- The authors report to have identified 195 loci significantly associated with five glyceic traits. Are these independent signals? I feel information are lacking in the results as to how to interpret the data.

We appreciate the reviewer's question.

First, we apologize for the initial oversight regarding the covariate for gestational week. In our analyses, we had used the OGTT week of gestation as the covariate for FPG and HbA1c. In the revised manuscript, we have corrected this to use the gestational week specific to FPG and HbA1c, respectively. Consequently, the lead SNP for FPG now ranges from 31 to 34, and for HbA1c, from 31 to 38, resulting in the identification of 205 independent lead SNPs for the five glyceic traits.

In the methods section, we clarified that we applied GCTA COJO analysis to select independent signals. Additionally, within the upper and lower 500k range, we

retained only the SNPs with the smallest p-value, ensuring the independence of these loci. In the revised manuscript, we have rephrased the first sentence of the “Large-scale genome-wide association study of GDM” and “Genome-wide association of five glycemic traits” sections as follows:

Lines 93-94: “We conducted a GWAS on 116,144 pregnant women, including 12,024 cases and 67,845 controls, identifying 19 independent loci significantly associated with GDM (Figure 3A).”

Lines 119 to 120: “Furthermore, we identified 205 independent loci significantly associated with five glycemic traits ($P < 5 \times 10^{-8}$), of which 115 were newly discovered (Figure 3B-E).”

Regarding the interpretation of the data, we have presented the key biological findings in Lines 257 to 276 of the revised manuscript.

5. The authors should address the power differences in T2D and GDM GWAS as some of the differences could be due to size differences between studies.

We thank the reviewer for raising this important issue. To address potential power differences between the T2D and GDM GWAS, we employed the ‘linemodls’ and ‘coloc’ approaches. The ‘linemodls’ R package (PMID: 36864614) probabilistically clusters explanatory variables based on effect sizes across multiple outcomes, considering scale, slopes, and correlation to account for sample size variability. We optimized scale parameters using an empirical Bayes method to minimize the impact of sample size differences on effect size estimates.

Additionally, to address shared genetic effects across datasets, we performed a colocalization analysis using the R ‘coloc’ package, which assesses overlap of individual SNPs in high LD between datasets via Bayesian testing. These methods collectively minimize the impact of power differences and improve the robustness of our results.

We have included the above explanation in the “Distinct and shared variants analysis” section of the Supplementary Notes.

6. Prediction models: As far as I can see the AUC confidence intervals between model 1-3 all overlap - how can the authors conclude that adding these genetics factors is actually better?

We thank the reviewer for raising this question. In the revised manuscript, we conducted a DeLong’s test to assess statistical differences in AUCs between the combined model and the phenotypic data model in the test dataset and the two

validation datasets. The results consistently showed statistically significant improvements in AUC for the combined model compared to the non-genetic model. Specifically, the combined model achieved AUCs of 0.729 and 0.710, with accuracies of 0.841 and 0.806 for predicting GDM before 20 weeks of gestation in the two validation datasets, significantly outperforming models relying solely on phenotypic data (DeLong's test $P < 0.05$), which achieved AUCs of 0.710 and 0.681, with accuracies of 0.841 and 0.807 in the first and second validation datasets, respectively.

Regarding the AUC confidence intervals, we apologize for any confusion in the presentation of our results. The reported AUC range in the manuscript is not a confidence interval but reflects the AUC of the parametric model across different data partitioning schemes during repeated 5-fold cross-validation. This approach was used to demonstrate the stability of our model. In the revised manuscript, we have clarified this point by rephrasing the paragraph (Lines 229 to 234) as follows:

*“In validation cohorts, the combined model achieved AUCs of 0.729 and 0.710 in the Baoan 20K and NIPT PLUS cohorts, respectively, with accuracy rates of 0.841 and 0.806. Compared to the phenotypic model, the combined model showed significance improvement in predictive performance, as demonstrated by DeLong's test (Baoan 20K validation: $P = 1.68 \times 10^{-2}$; NIPT PLUS validation: $P = 4.54 \times 10^{-5}$; **Table S17**).”*

Reviewer #3 (Remarks to the Author)

This is a manuscript by Yuqin Gu et al. reporting a study that used non-invasive prenatal screening to build PRS and combined with biomarkers measured <20 weeks of gestation for prediction of GDM at 24-28 weeks in large population of pregnant women in China.

The study has multiple strengths, including large sample size, leveraging two hospital-based samples, with similar rate of GDM (14-15%), using NIPT genotyping data + 43 biomarkers, performing GWAS of multiple glucose traits in addition to GDM diagnosis, and used of recent and appropriate statistical approaches for genetic analyses (e.g. for building PRS, and for comparison of effect size for pregnancy versus outside of pregnancy)

My concerns are mostly regarding interpretation and conclusions put forward by the authors. First, they should acknowledge that this approach may not be applicable to other settings where NIPT does not offer standard genome sequencing or large-scale genotyping. Second, their stated conclusions are a bit strong, the most informative marker is still glucose – in this current study measured at 16 weeks, to predict GDM at 24 weeks (yes, glucose predicts glucose).

We thank the reviewer for the insightful comments and suggestions. We acknowledge the reviewer's concern that our method may not be applicable to hospital settings where

NIPT was not provided as a standard screening approach for fetal trisomies. However, we would like to note that since the NEXT randomized controlled trials in 2015 (PMID: 25830321), NIPT has been recommended as a first-tier screening tool for trisomy prevention in numerous countries, including the Netherlands, Belgium, the United Kingdom, Iceland, China, Australia, Singapore, Hong Kong, and others. As a result, NIPT has become widely adopted in clinical practice globally. In the revised manuscript, we have added a discussion of this limitations in the “Limitation of the study” section (Lines 328 to 334) as follows:

“Finally, our method may have limited applicability in hospital settings where NIPT is not routinely offered as a standard screening tool for fetal trisomies. Nevertheless, since the publication of the NEXT RCT study in 2015²⁹, NIPT has increasingly been recommended as a first-tier screening tool for trisomy prevention in many countries and is now widely adopted in clinical practice worldwide. The incorporation of genetic data into GDM prevention and treatment strategies could be particularly valuable in these settings. ”

Regarding the reviewer’s concerns about the strength of our conclusion over the contribution of PRS to GDM prediction, we conducted a DeLong’s test to evaluate statistical differences in AUCs. Our results showed statistically significant differences in AUC between the combined model and the non-genetic model across the test, validation, and external validation datasets, indicating the importance of incorporating genetic factors in prediction models.

However, we fully recognize that glucose remains an important factor in early GDM prediction. In the revised manuscript, we have addressed this in the discussion section (Lines 279 to 290) as follows:

“In this study, the combined model achieved an AUC of 0.710 and 0.729, and an accuracy of 0.841 and 0.806 for predicting GDM before 20 weeks of gestation in the two validation datasets, significantly outperforming models relying solely on phenotypic data (DeLong’s test $P < 0.05$), which achieved an AUC of 0.710 and 0.681 and an accuracy of 0.841 and 0.807 in the first and second validation datasets, respectively. Our explainable tree-based models identified key predictive factors for GDM, such as gestational age, BMI and FPG levels in early pregnancy. These findings align with previous predictive model based on clinical features¹⁰, reinforcing the importance of early glucose levels in GDM prediction. Notably, PRS for GDM, OGTT2H and OGTT1H emerged as crucial predictors, highlighting the value of integrating genetic information into GDM prediction models.”

I also have a few additional specific comments below:

Methods:

How were excluded patients with pre-gestational diabetes mellitus (PGDM)? Were A1c at 12 weeks, and/or FPG at 16 weeks used to 'exclude' pre-gestational diabetes? If so, what thresholds were used?

We thank the reviewer for raising this question. During the data collection process, we gathered clinicians' diagnostic records, and participants diagnosed with pre-gestational diabetes mellitus (PGDM) were excluded from the GDM diagnostic process (Methods section, Lines 368 to 372). The diagnosis of PGDM was made based on the clinicians' inquiry into the participants' diabetes disease history. In our analysis, we did not use HbA1c at 12 weeks or FPG at 16 weeks to exclude PGDM, in accordance with the GDM diagnosis criteria (PMID: 20190296).

The authors should be clearer in their methods about the difference between fasting glucose (I think meaning measured at 16 weeks of gestation) and glucose at time = 0 for the OGTT (which is also fasting but measured at 24-28 weeks).

We thank the reviewer for raising this question. The reviewer's understanding is correct: FPG level were measured between gestational weeks 16 to 18, while OGTT0H was assessed at time=0 using a 75g oral glucose tolerance test (OGTT) between 24 to 28 weeks of gestation. In the revised manuscript, we have clarified this distinction in the following rephrased paragraph (lines 146 to 149):

"The genetic determinants of FPG (examined at ~16 gestational week) and OGTT0H (measured between 24 and 28 gestational week) were highly similar, sharing 22 significant loci. The lead SNP effect sizes were strongly correlated (Pearson's $r = 0.975$, $P < 2.2 \times 10^{-16}$), with a genetic correlation of $0.943(0.026)$, $P = 9.33 \times 10^{-289}$ (Figure 3B-C, Figure S11 , Table S8)."

In figure 1 and figure 5, can the authors include the population (sample size) used of the predictive analyses (XGBoost)

We thank the reviewer for providing valuable suggestions. During the revision, we have clarified the sample size for evaluating the model performance in the manuscript. In Figure 1 and Figure 5 (see below), as well as in the Figure 2, we present the model performance using the test dataset, which consists of 2,826 samples (550 GDM cases and 2,276 controls). For the two validation cohorts, the Baoan 20K cohort includes 523 cases and 2,303 controls, while the NIPT PLUS cohort includes 795 cases and 2,770 controls. These sample sizes are detailed in Figure 2. The AUC and accuracy for the two validation cohorts are similar to those observed in the test dataset, suggesting the robustness of the prediction model.

[figure redacted]

Figure 1

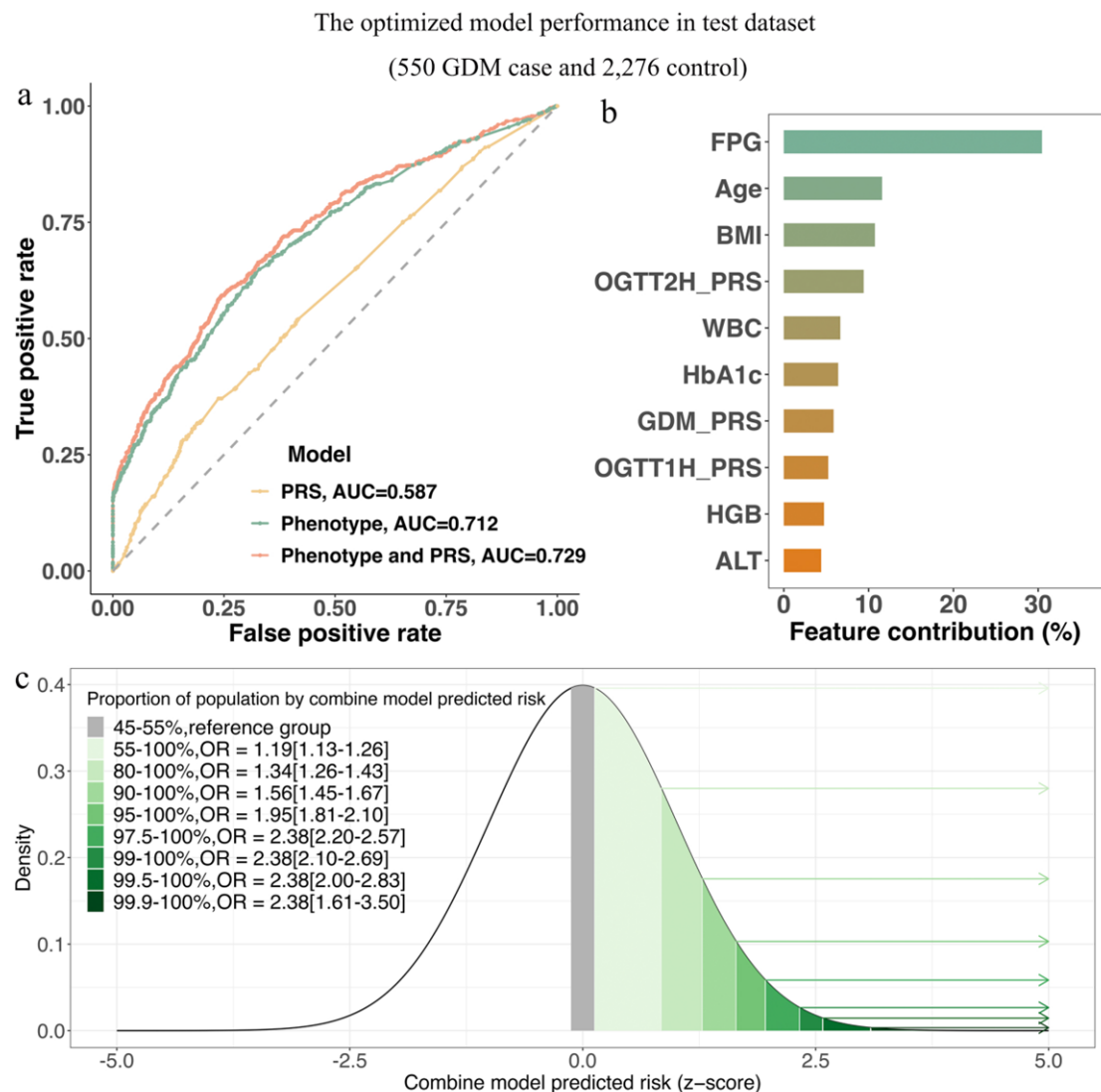


Figure 5

Results:

The authors stated that the fasting glucose was measured in first trimester (line 240) but in Supp Table 1 – it's indicated mean gestational age 16 weeks.

We thank the reviewer for raising these questions. After your reminder, we reconfirmed the fasting glucose measured time in this study, the mean FPG measured week was 16.72 as supplementary Table S11 displayed, which was early to middle stage of pregnancy. In the revised manuscript, we have rephrased the paragraph as follows (lines 146 to 149):

“The genetic determinants of FPG (examined at ~16 gestational week) and OGTT0H (measured between 24 and 28 gestational week) were highly similar, sharing 22

significant loci. The lead SNP effect sizes were strongly correlated (Pearson's $r = 0.975$, $P < 2.2 \times 10^{-16}$), with a genetic correlation of $0.943(0.026)$, $P = 9.33 \times 10^{-289}$ (Figure 3B-C, Figure S11, Table S8).”

GDM and T2D in East-Asian: the authors found that *MTNR1B* is shared but the visual inspection of their own Figure 4A (and prior knowledge from European GDM versus T2D) suggest GDM predominant – how do the authors explained the discrepancies?

We thank the reviewer for raising this question. In the previous version of our manuscript, we identified *MTNR1B* as shared between GDM and T2D because the colocalization PPH4 for this locus was greater than 0.8.

Upon further review, prompted by your comment, we examined the LocusZoom plot for *MTNR1B* in both GDM and Asian female T2D, and found that the most significant locus for *MTNR1B* in Asian female T2D was rs10830963 ($P=3.88 \times 10^{-5}$) (Figure R6). We then revisited our colocalization analysis and found that we had included 4,000-5,000 SNPs in one region, which is much higher than the recommended 1,000-2,000 SNPs suggested by the ‘coloc’ method. This likely compromised the reliability of our results. To address this, we adjusted the region length from 1M to 500kb, resulting in a revised PPH4 of 0.468 for the colocalization analysis, compared to the original PPH4>0.8 for this locus.

Besides the revision for *MTNR1B*, the PPH4 for *GCKR* (lead SNP rs11127048) was revised from 0.545 to 0.193. Therefore, we updated our conclusion regarding *MTNR1B* and *GCKR*, shifting the classification from shared between GDM and T2D to GDM predominant.

In the revised manuscript, we have rephrased the conclusion as follows (lines 177-179):
“Consistent with findings from European studies²², the *GCKR*, *PCSK1* and *MTNR1B* loci demonstrated a GDM-predominant effect.”

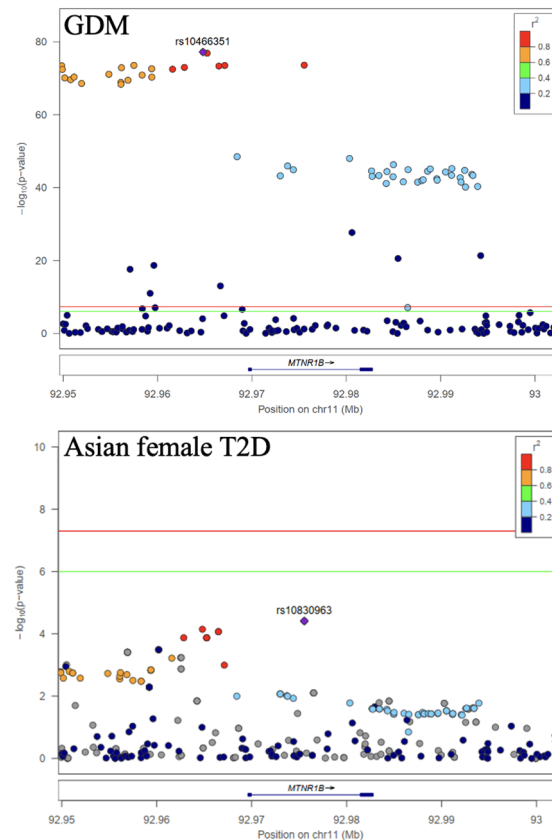


Figure R6. LocusZoom plot of *MTNR1B* in GDM and Asian female T2D GWAS results.

In line 294: I think the authors meant Table S1:
and 43 biomarkers and electronic health record phenotypes listed in Table S2

We thank the reviewer for raising this issue and apologize for the oversight. In the revised manuscript, we have relocated Table S1 to Table S12 and corrected the typo in Line 210, as follows:

“We used the PRSice-2 algorithm²⁴ and PLINK(version 2.0) to construct the PRS models, with details of P-value thresholds, SNP counts, and R^2 values provided in Table S12.”

In lines 300-302: please add the gestational weeks at which the FPG and A1c was measured for clarity. The first model using 46 clinical features achieved an AUC of 0.700 [0.643-0.745], and an accuracy of 0.821[0.803-0.827] (Figure 5A, Table S8), with FPG (11.8%), maternal age (6.57%), and HbA1c (4.33%) as the top contributors (Figure S8)

We thank the reviewer for raising this question.

In the revise manuscript, we have rephrased the paragraph as follows (Lines 219 to 222):

“The non-genetic model, based on 46 clinical features, achieved an AUC of 0.712, and an accuracy of 0.835 in the test dataset (Figure 5A, Table S13). Key contributors were FPG (measured at ~16 gestational weeks; 38.12%), maternal age (14.03%), and BMI (measured at ~16 gestational weeks, 11.39%) (Figure S12, Table S14).”

Discussion: the authors mis-reported their own findings in lines 350-352

In Figure 5, it's the PRS model that has an AUC =0.630 (not the phenotypic model)

From the Discussion text: This combined model achieved an AUC of 0.758[0.682-0.768], and an accuracy rate of 0.830 [0.807-0.837] before 20 weeks of gestation, significantly outperforming models using only phenotypic data, which had with an AUC of 0.630 [0.559-0.654] and an accuracy rate of 0.803 [0.772-0.810].

We thank the reviewer for raising this question. We apologize for the error in our previous version of the manuscript. In the revise manuscript, we have corrected the paragraph as follows (lines 279 to 284):

“In this study, the combined model achieved an AUC of 0.710 and 0.729, and an accuracy of 0.841 and 0.806 for predicting GDM before 20 weeks of gestation in the two validation datasets, significantly outperforming models relying solely on phenotypic data (DeLong’s test $P < 0.05$), which achieved an AUC of 0.710 and 0.681 and an accuracy of 0.841 and 0.807 in the first and second validation datasets, respectively.”

Discussion: could the authors expand about how they would use the predicting algorithm? If they identify the 0.1% of pregnant women at ‘high risk’ of GDM (based on the cut-off for OR at 1.65) would they target these women by asking to do an OGTT earlier than the standard 24 weeks? Or offer them right away monitoring blood glucose and treatment?

We appreciate the editor for raising this concern. In our view, individuals identified as "high risk" for GDM should be considered at increased risk of developing the disease. As medically appropriate, enhanced monitoring and the implementation of intervention strategies, informed by clinical experience, are recommended. However, the lack of established medical guidelines complicates the widespread adoption of these measures, and their effective implementation will require coordinated efforts across multiple sectors to support early prevention and intervention of GDM.

A recent randomized controlled trial (RCT), published in *The New England Journal of Medicine* in 2023, demonstrated the benefits of early management of GDM before 20 weeks, which included education, dietary counseling, and instructions on monitoring capillary blood glucose levels. These findings serve as important references for shaping our future intervention strategies.

In the revise manuscript, we have rephrased the discussion as follows (Lines 301 to 303):

“For individuals identified as at high risk of developing GDM, it is recommended that monitoring be increased and intervention strategies, informed by clinical experience, be implemented as early as medically permissible.”

Manuscript Title: Genetic Architecture and Risk Prediction of Gestational Diabetes Mellitus in over 116,144 Chinese Pregnancies

Reviewer Comments & Author Rebuttals

We would like to express our gratitude to the reviewers for their constructive feedback and valuable recommendations. We have carefully addressed the remaining questions, mainly from reviewer 2 and present our point-by-point responses below. The revised contents are highlighted in blue text in the revised manuscript, and the changes are quoted in italics within this response.

Author Rebuttals to Initial Comments:

Referees' comments:

Referee #1 (Remarks to the Author):

All of my comments have been addressed to my satisfaction.

Reviewer #2 (Remarks to the Author):

I appreciate the authors for their response to my concerns. I am however not quite satisfied with some of their responses.

1. I suggested using a software like REGENIE or BOLT-LMM to handle population stratification in the sample. Just removing related individuals is not enough to mitigate the effect of population stratification. The authors response is with concerns regarding SAIGE. I have myself seen these issues with SAIGE and would not recommend this software, however, the field standard is to use a software with a GRM when that is possible, and I do think the authors should reconsider using a best practice tool for their analysis.

We sincerely appreciate the reviewer's suggestion regarding the choice of GWAS methodology to account for population stratification. In response, we conducted additional GWAS analyses using REGENIE (v4.1) and BOLT-LMM (v2.4.1) for GDM and five glycemic traits included in our study.

First, the GWAS results obtained using REGENIE and BOLT-LMM were highly consistent with those from PLINK2's regression model, with *Pearson's R* = 1.00 across all phenotypes (Figures R1 and R2; also presented in the revised manuscript as Figures S1, S2, and Table S2).

Second, when evaluating GWAS results using LDSC intercepts and ratios, PLINK2 provided values closest to the expected 1 for intercepts and 0 for ratios for FPG, OGTT1H, and OGTT2H in the highest proportion of phenotypes. REGENIE produced the most appropriate estimates for HbA1c and OGTT0H, while BOLT-LMM yielded the most suitable results for GDM (Table R1, corresponding to Table S3 in the revised manuscript). This consistency across methods suggests that cryptic relatedness is unlikely to be a major concern in our dataset, particularly given that

participants were recruited within a five-year period and duplicate samples—namely, pregnant women with a second child—were excluded (Methods, Lines 350–352). However, we acknowledge that as sample sizes increase, this assumption may require reassessment.

Based on these findings, we have chosen to retain PLINK2 as the primary analysis method in the manuscript. Nevertheless, to address the reviewer’s concern—one that is also relevant to the broader audience—we have incorporated the comparison results for PLINK2, REGENIE, and BOLT-LMM as supplementary material.

In the revised manuscript, we have included the following updates:

Results section (Lines 95–101):

"We conducted a GWAS on 116,144 pregnant women, including 12,024 cases and 67,845 controls, using three analytical methods: PLINK2, REGENIE, and BOLT-LMM (Methods). The results indicate that genetic effect estimates for the lead SNPs are highly consistent across methods (Figures S1–S2, Table S2). Among the three approaches, PLINK2 provided the expected LDSC intercepts and ratios for the highest proportion of phenotypes (Table S3). Therefore, we selected PLINK2 as the primary method for reporting results."

Methods section (Lines 409–414):

"We also compared the GWAS results obtained using PLINK2 with those from REGENIE (v4.1) and BOLT-LMM (v2.4.1), which demonstrated high consistency across methods. This consistency may be attributed to the absence of cryptic family relatedness in the dataset. Since PLINK2 produced the expected intercepts and ratios for the highest proportions of phenotypes, we selected it as the primary method for reporting results."

We hope this revision addresses the reviewer’s concerns and clarifies our rationale for selecting PLINK2 while providing transparency regarding alternative analyses.

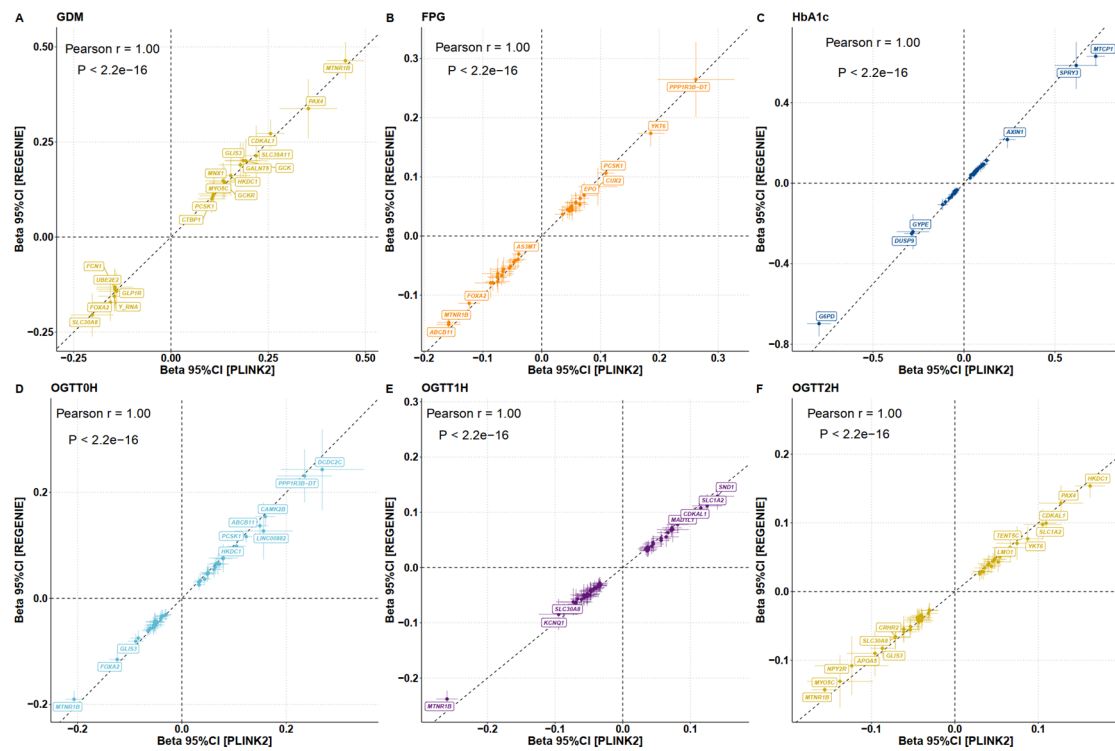


Figure R1. Comparison of associations for lead SNPs in GDM and five glycemic traits between PLINK2 and REGENIE.

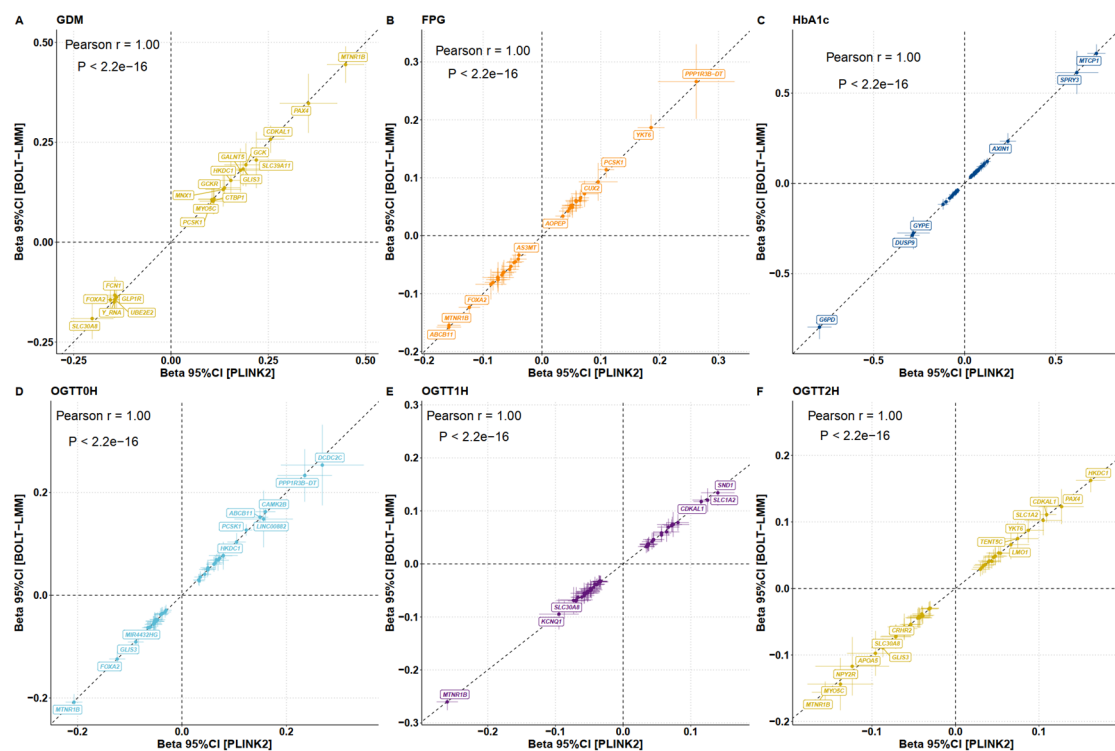


Figure R2. Comparison of associations for lead SNPs in GDM and five glycemic traits between PLINK2 and BOLT-LMM.

Table R1. LDSC estimates of lambda GC, intercept, and ratio for GWAS results obtained using PLINK2, REGENIE, and BOLT-LMM, respectively

Trait	Total_Observed_scale_h2	Lambda_GC	Mean_Chi_2	Intercept	Ratio
FPG_plink	0.1389(0.0148)	1.1082	1.1631	1.0154(0.0042)	0.0947(0.0256)
FPG_regenie	0.118(0.0153)	1.0864	1.1444	1.0161(0.0046)	0.1117(0.0322)
FPG_bolt_lmm	0.1266(0.0163)	1.0957	1.1556	1.0195(0.0048)	0.1255(0.0307)
GDM_plink	0.0692(0.0071)	1.1019	1.1277	1.0233(0.0035)	0.1822(0.0273)
GDM_regenie	0.0661(0.007)	1.1428	1.1673	1.0661(0.0041)	0.3951(0.0244)
GDM_bolt_lmm	0.0705(0.0074)	1.1019	1.1295	1.0220(0.0040)	0.1699(0.0309)
HbA1c_plink	0.1117(0.0094)	1.1144	1.1691	1.0257(0.0032)	0.152(0.0187)
HbA1c_regenie	0.0915(0.009)	1.0988	1.1452	1.0228(0.0033)	0.1574(0.0227)
HbA1c_bolt_lmm	0.1038(0.0098)	1.1082	1.1629	1.0271(0.0034)	0.1661(0.021)
OGTT0H_plink	0.132(0.0151)	1.1113	1.2305	1.0247(0.0057)	0.107(0.0249)
OGTT0H_regenie	0.1211(0.0151)	1.1019	1.214	1.0214(0.0063)	0.1001(0.0293)
OGTT0H_bolt_lmm	0.132(0.0169)	1.1082	1.2338	1.0258(0.0068)	0.1105(0.029)
OGTT1H_plink	0.1701(0.0134)	1.1973	1.2946	1.0327(0.0065)	0.111(0.0221)
OGTT1H_regenie	0.1439(0.0116)	1.1683	1.2621	1.0345(0.0068)	0.1315(0.026)
OGTT1H_bolt_lmm	0.1573(0.0127)	1.1811	1.2817	1.0345(0.0075)	0.1223(0.0267)
OGTT2H_plink	0.1241(0.0102)	1.1428	1.217	1.0253(0.0043)	0.1165(0.0199)
OGTT2H_regenie	0.113(0.0093)	1.1396	1.2086	1.0291(0.0046)	0.1397(0.0221)
OGTT2H_bolt_lmm	0.1183(0.0098)	1.1333	1.2155	1.0304(0.0049)	0.1412(0.0229)

Bolded lines indicate the results with the lowest statistical inflation, where the LDSC intercept is closest to 1 and the ratio is closest to 0, compared across PLINK2, REGENIE, and BOLT-LMM.

- The authors have changed their reporting of new loci to state that they identified 19 independent loci. To me it is still not clear from their description in the methods how many of these are novel. Have they been associated with GDM in other populations? Or Type 2 Diabetes? The authors state that they were not found in "European GDM GWAS finding", but the paper is not referenced and there are several papers published that should be checked. Further on this they state that 205 independent signals were associated with the five glycemic traits. I still wonder if it is in fact 200 new loci or is it 40 loci in each of the 5 GWAS? And are they novel? Or previously known GDM, glucose or T2D hits?

We appreciate the reviewer's insightful comments and apologize for any lack of clarity in our original description of locus novelty. In response, we have thoroughly revised the relevant sections of the manuscript to explicitly define novel loci and provide additional references to support our claims.

In the revised manuscript, we have clarified the definition of novel loci in a newly added subsection of the Methods section (Lines 416 to 431) as follows:

"Independent loci consisting of SNPs located within 500 kb upstream or downstream of a lead SNP previously associated with the same phenotype (i.e.,

GDM or any of the five glycemic traits) were classified as known loci, while all others were considered novel loci. Previous genetic association findings were determined based on the latest version of the GWAS Catalog, released on February 18, 2025 (gwas_catalog_v1.0.2-associations_e113_r2025-02-18.tsv).

Specifically, for GDM, novel loci were defined as those that did not contain any SNPs previously reported in genetic association studies within a 1Mbp window centering the lead SNP. We did not consider comparisons with T2D in defining novel loci for GDM.

For glycemic traits, novel loci were defined as those not previously associated with the same glycemic trait, regardless of whether the association was identified in pregnant or non-pregnant individuals. Comparisons with T2D or GDM were not considered in defining novel loci for glycemic traits”.

Specifically, to ensure accuracy, we have updated our analysis using the latest GWAS Catalog file (gwas_catalog_v1.0.2-associations_e113, released on 2025-02-18), replacing the previously used version (gwas_catalog_v1.0.2-associations_e109). Based on this updated dataset, we revised the classification of certain loci, resulting in minor changes to our statistics on novel loci. Specifically, the following loci were reclassified as known loci: GDM (*SLC30A8* rs9650069), FPG (*PSMD3* rs3907022), HbA1c (*SBF2* rs11042493, *LINGO1* rs12901130), and OGTT2H (*CDKAL1* rs35261542). The remainder of the results remain unchanged. Accordingly, we have specified the use of the latest GWAS Catalog file in the Methods section and marked the update to certain statistics in the GWAS results section in blue text in the revised manuscript.

Regarding GDM loci, we identified 19 independent loci associated with GDM, of which 13 were novel based on our defined criteria. These novel loci represent the first reported associations with GDM globally.

To address the reviewer’s concern regarding European GDM GWAS findings, we have incorporated the citation to the largest European GDM GWAS to date, conducted in the FinnGen cohort (12,332 cases and 131,109 controls, PMID: 38182742). This study identified 13 GDM-associated loci, which were included in the gwas_catalog_v1.0.2-associations_e113_r2025-02-18.tsv file. Therefore, our classification of GDM novel loci have excluded these previously identified loci.

We have clarified this point in the revised manuscript (Result section, Lines 108 to 110) as follows: “*Notably, thirteen loci were identified for the first time, even when compared to recent European GDM GWAS findings¹⁸ (highlighted in red, Figure 3A; Table S4).*”

For glycemic traits, we identified a total of 205 independent loci significantly associated with at least one of the five glycemic traits: 34 for FPG, 42 for OGTT0H, 50 for OGTT1H, 41 for OGTT2H, and 38 for HbA1c. Among these, 111 loci were novel, including 11 for FPG, 13 for OGTT0H, 43 for OGTT1H, 34 for OGTT2H, and 10 for HbA1c.

In response to the reviewer's question regarding whether the 205 loci represent unique loci or overlapping signals across multiple traits, we clarify that the association findings are phenotype-specific. We did not merge loci across different traits, and thus the 205 loci, including the 111 novel loci reflect distinct signals for each trait rather than a cumulative, non-overlapping total across all five glycemic traits.

We have clarified this point in the revised manuscript (Lines 127 to 130) as follows: *"Furthermore, we identified 205 independent loci significantly associated with five glycemic traits ($P < 5 \times 10^{-8}$): 34 for FPG, 42 for OGTT0H, 50 for OGTT1H, 41 for OGTT2H and 38 for HbA1c. Of these, 111 loci were newly discovered, with 11 associated with FPG, 13 with OGTT0H, 43 with OGTT1H, 34 with OGTT2H and 10 with HbA1c (Figure 3B-E, Table S4)."*

We appreciate the reviewer's feedback, which has helped us improve the clarity and rigor of our manuscript. We hope this revisions adequately address the reviewer's concerns.

Reviewer #3 (Remarks to the Author)

The authors have addressed my comments appropriately.

Manuscript Title: Genetic Architecture and Risk Prediction of Gestational Diabetes Mellitus in over 116,144 Chinese Pregnancies

Reviewer Comments & Author Rebuttals

We would like to express our gratitude to the reviewers for their constructive feedback and valuable recommendations. We have carefully addressed the remaining question, from reviewer 2 and present our point-by-point responses below. The revised contents are highlighted in blue text in the revised manuscript, and the changes are quoted in italics within this response.

Author Rebuttals to Initial Comments:

Referees' comments:

Reviewer #2 (Remarks to the Author):

I thank the authors for taking onboard the comments and addressing some of them in a rigorous way. I still don't 100% agree with their rationale for using plink2, but I am glad they have shown that it for the main is showing concordant results as for BOLT or REGENIE.

For my second point of my last review I stated that I am not clear on how many findings are actually novel, and I still am not based on the response from the authors. To me it reads as inflation of the results because the authors have performed 5 GWAS of glycemic traits in addition to GDM. All these traits are highly correlated and when I look at table S4 which list all the SNPs and their genes I see that many of the genes are present in more than one GWAS. For example variants in CDKAL1, PCSK1, MTNR1B to name are few are listed over and over again. It sounds very impressive when the authors write 205 loci, but reality is seemingly a lot fewer new genes have been associated with any of these traits.

The authors write in the response: "we clarify that the association findings are phenotype-specific. We did not merge loci across different traits" and I wonder why you would not consider Fasting glucose at tow time points to be the same or similar phenotype and cross check if the same gene pop up in both these GWASs?

To me this reporting feels inflated and not accurate, and perhaps this is also reflected in the title quoting an impressive "116,144 Chinese Pregnancies" when the individual GWASs have a sample size ranging from 56,000 to 85,000?

We sincerely thank the reviewer for the thoughtful comments and critical feedback.

We acknowledge the reviewer's concern regarding potential overstatement in reporting novel loci across multiple glycemic traits. In response, we have re-evaluated our

findings by treating the five glycemic traits as a composite phenotype. As reported in the revised manuscript (Lines 136–138), this approach yielded 140 genome-wide significant loci, of which 79 were novel.

Regarding the reviewer's comment on locus overlap, we would like to clarify that our primary analyses were trait-specific, and we did not merge loci across different glycemic traits to avoid conflating signals from correlated phenotypes. However, we agree that variants associated with multiple traits—such as those near CDKAL1, PCSK1, and MTNR1B—represent shared genetic architecture rather than independent findings. We now explicitly acknowledge this in the manuscript to ensure transparency and to avoid the impression of inflation.

We have revised the Results section (Lines 136–138) to clarify this point and better reflect the number of distinct loci identified across traits as follows:

“When analyzing the five blood glycemic traits as a composite phenotype and consolidating overlapping loci, we still identified 140 genome-wide significant association loci, of which 79 were novel.”

As for the sample size noted in the title, we appreciate the reviewer's suggestion and understand the concern. While our full dataset includes 116,144 participants with NIPT-based biochemical and glycemic measurements, the sample sizes for individual GWAS analyses varied (ranging from ~56,000 to ~85,000) due to differences in available phenotype data. To address this, we have removed the sample size from the title. The revised title now reads:

"Genetic Architecture and Risk Prediction of Gestational Diabetes Mellitus in Chinese Pregnancies."

We hope these revisions clarify our findings and address the reviewer's concerns.